

Diffusive Alignment of Heterogeneous Features for Visual Classification

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Abstract

Cross-modal alignment enhances visual representations by introducing discriminative textual semantics. However, existing methods usually rely on unidirectional or implicit alignment mechanisms, which are insufficient to model complex semantic relationships under large modality gaps and heterogeneous feature distributions. To address this issue, we propose **Reliability-Consistency Guided Diffusion Alignment**, termed *RCDAlign*, a diffusion-based framework for robust heterogeneous visual-semantic feature alignment. Specifically, Conditional Feature Diffusion progressively guides visual features toward a high-quality semantic target space through a conditional denoising process. Reliability-Consistency Adaptive Fusion jointly estimates textual reliability and visual-textual consistency to suppress noisy or inconsistent privileged guidance during cross-modal interaction. Furthermore, Confidence-Guided Selection adaptively controls the contribution of fused semantic features according to feature divergence and classification confidence, thereby alleviating negative transfer caused by semantic conflicts in challenging samples. Experiments on three public datasets demonstrate that RCDAlign improves the discriminability of visual representations and achieves competitive performance against existing alignment methods under diverse visual-semantic scenarios.

CCS Concepts

• **Computing methodologies** → *Image representations; Visual content-based indexing and retrieval; Markov decision processes.*

Keywords

Cross-modal alignment, Diffusion model, Classification.

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1 Introduction

Visual representation quality is crucial for accurate visual understanding and classification. However, variations in illumination, viewpoint, background, and object appearance make it difficult to learn robust semantic features from visual data [43]. To address this issue, existing studies have explored multi-modal alignment, which

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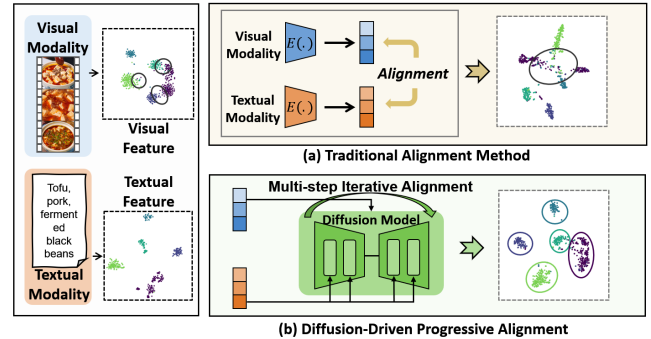


Figure 1: Motivation of the proposed RCDAlign. Direct one-step alignment is vulnerable to noisy or inconsistent textual guidance, while progressive diffusion-based alignment gradually refines the visual feature toward reliable semantic targets.

leverages complementary information from different modalities and uses textual semantics to improve visual feature distributions [33]. Despite its effectiveness, cross-modal alignment can be limited by semantic inconsistency and high heterogeneity between modalities, as visual and textual representations often differ substantially in structure, semantics, and distribution, as illustrated in Fig. 1.

Existing alignment methods can be broadly divided into global feature alignment and local region alignment. Global approaches typically perform direct feature mapping [27, 39, 40] or implicit semantic fusion [4, 42], while local methods extract regional representations through grids, patches, or semantic segmentation [35, 49], and enhance cross-modal interactions using attention mechanisms, contrastive learning, or graph-based modeling [44, 55]. Beyond discriminative alignment, generative models further model the joint distribution of visual and textual modalities [42]. Large-scale contrastive vision-language pretraining, such as CLIP [37], has also demonstrated that aligning visual and textual embeddings at scale can yield transferable representations, motivating subsequent knowledge-distillation-based alignment approaches [13, 54]. In particular, diffusion models [20] have shown strong generative ability by learning a progressive denoising process, and this generative paradigm has recently been extended to cross-modal discrimination [21] and text-video retrieval [25]. However, most existing cross-modal diffusion methods are primarily generation-oriented and rely on implicit interactions such as self-attention or cross-attention, without explicitly modeling the alignment process. This makes them vulnerable to noise and semantic mismatch, while direct alignment between discrete textual symbols and continuous visual representations may also disturb the intrinsic structure of each modality.

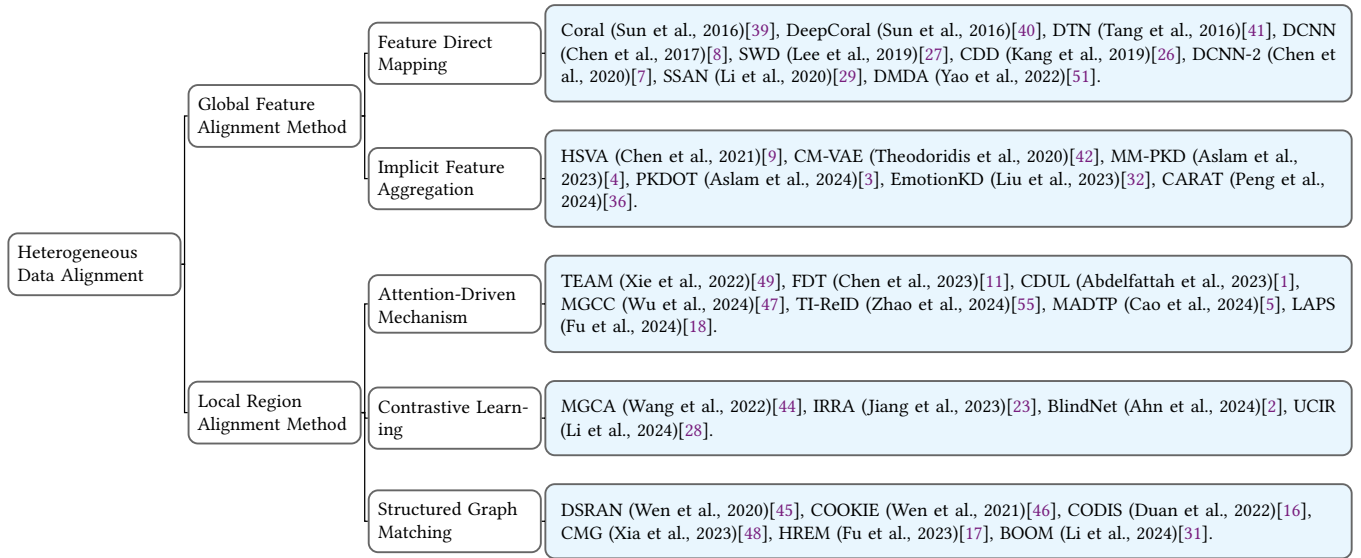


Figure 2: Taxonomy for Heterogeneous Data Alignment.

To address these limitations, we propose Reliability-Consistency Guided Diffusion Alignment (RCDAlign), a diffusion-based framework for heterogeneous visual-semantic feature alignment. RCDAlign formulates visual-to-textual alignment as a Conditional Feature Diffusion (CFD) process, where textual features define the semantic target distribution and visual features provide conditional guidance. To reduce unreliable cross-modal guidance, Reliability-Consistency Adaptive Fusion (RCAF) jointly models textual reliability and visual-textual consistency. Furthermore, Confidence-Guided Selection (CGS) adaptively controls the contribution of fused semantic features according to feature divergence and classification confidence, thereby alleviating negative transfer. The main contributions are summarized as follows:

- We propose RCDAlign, a reliability-consistency guided diffusion alignment framework that reformulates heterogeneous feature alignment as a conditional denoising process.
- We design RCAF to adaptively suppress noisy or inconsistent textual guidance by jointly considering semantic reliability and cross-modal consistency.
- We introduce CGS to select reliable fused representations based on feature divergence and prediction confidence, reducing negative transfer from unreliable privileged information.

2 Literature Review

Heterogeneous data alignment methods can be broadly categorized into global feature alignment methods and local region alignment methods, as summarized in Fig. 2. Global methods operate at the holistic level through feature direct mapping or implicit feature aggregation, whereas local methods establish fine-grained correspondence via attention-driven mechanisms, contrastive learning, or structured graph matching. We discuss each category below.

2.1 Global Feature Alignment Methods

Global feature alignment aims to learn modality-consistent representations between visual and textual features at the holistic level. Existing methods can be broadly divided into feature direct mapping and implicit feature aggregation. Direct mapping methods align heterogeneous modalities by projecting features into a target or shared representation space. For example, CORAL [39] and Deep CORAL [40] reduce domain discrepancy by minimizing covariance-based feature differences, while SSAN [29] and CDD [26] further introduce class-aware alignment to reduce intra-class variation and enlarge inter-class separation.

Implicit feature aggregation methods perform alignment as an intermediate step during multimodal interaction or reconstruction. HSVA [9] maps visual and semantic features into a structurally aligned common space through task-specific encoders and adversarial discrepancy learning. CARAT [36] enhances cross-modal complementarity by reconstructing and fusing modality-specific information. CM-VAE [42] employs variational autoencoders to learn latent cross-modal mappings through an intermediate distribution. In a related line of work, knowledge-distillation-based alignment methods, such as DMD [30], C2KD [22], and Emo-DNA [52], implicitly transfer privileged semantic knowledge from a richer modality to a target modality, while SCPNet [14] exploits structured semantic priors to guide implicit feature aggregation under incomplete supervision. Although these methods improve global consistency, they rely on direct or implicit one-step alignment, which may be insufficient when modality gaps and semantic noise exist.

2.2 Local Region Alignment Methods

Local region alignment focuses on fine-grained correspondence between image regions and textual fragments. These methods usually extract regional visual representations through grids, patches, or semantic regions, and can be grouped into attention-based alignment, contrastive learning, and graph-based matching. Attention-based

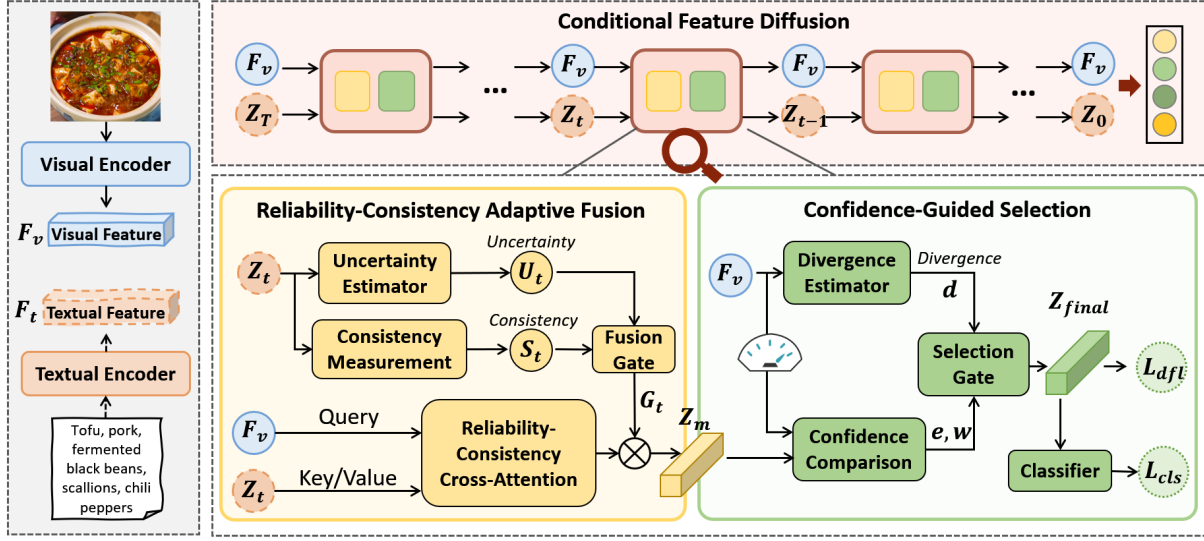


Figure 3: Overview of the proposed RCDAlign framework.

methods dynamically model cross-modal associations between local units. TEAM [49] performs both global-level and token-level alignment, while LAPS [18] selects semantically salient visual patches and calibrates their semantic representations. MGCC [47] combines attention and contrastive learning to capture key image-text correspondences at both coarse and fine levels.

Contrastive learning methods construct positive and negative cross-modal pairs to pull matched representations closer and push unmatched ones apart. IRRRA [23] enhances cross-modal consistency by optimizing the similarity between positive visual-textual pairs, while MGCA [44] performs contrastive alignment at instance, token, and semantic levels. In addition, graph-based methods explicitly model structural relationships between image regions and text units. COOKIE [46], CODIS [16], and HREM [17] build cross-modal relational structures to capture local semantic dependencies and improve image-text matching. Despite their effectiveness, local alignment methods often require complex region-level interactions and may still suffer from unreliable textual guidance or semantic inconsistency.

3 Problem Formulation

Let the training set be represented as

$$\mathcal{D}_{train} = \{(x_i^v, x_i^t, y_i)\}_{i=1}^N, \quad (1)$$

where x_i^v denotes the visual input, x_i^t denotes the paired textual description, and y_i is the corresponding class label. The textual modality is treated as privileged information, which is available only during training.

At test time, only visual samples are provided:

$$\mathcal{D}_{test} = \{x_j^v\}_{j=1}^M. \quad (2)$$

A visual encoder $f_v(\cdot)$ and a text encoder $f_t(\cdot)$ are used to extract visual and textual representations:

$$v_i = f_v(x_i^v), \quad z_i = f_t(x_i^t), \quad (3)$$

where $v_i \in \mathbb{R}^d$ and $z_i \in \mathbb{R}^d$. The objective is to learn a conditional alignment model that generates a semantically enhanced feature $\hat{z}_i$ from the visual feature v_i , and then predicts the class label through a classifier $C(\cdot)$:

$$\hat{y}_i = C(\hat{z}_i). \quad (4)$$

4 Method

We propose Reliability-Consistency Guided Diffusion Alignment (RCDAlign), which contains three modules: Conditional Feature Diffusion (CFD), Reliability-Consistency Adaptive Fusion (RCAF), and Confidence-Guided Selection (CGS). The overall architecture is shown in Fig. 3. Given an image-text pair, the visual and textual encoders extract token-level representations $F_v \in \mathbb{R}^{N_v \times D}$ and $F_t \in \mathbb{R}^{N_t \times D}$, where N_v and N_t denote the numbers of visual and textual tokens. For notational simplicity, F_v and F_t correspond to the encoder outputs v_i and z_i defined in Sec. 3, and the output Z_{final} of CGS is the enhanced feature $\hat{z}_i$ fed into the classifier $C(\cdot)$. RCDAlign aims to progressively align visual features with textual semantics while suppressing unreliable privileged guidance.

4.1 Conditional Feature Diffusion

To align visual representations with high-level textual semantics, we formulate feature alignment as a conditional denoising process. The textual feature F_t is treated as the clean semantic target, while the visual feature F_v serves as the condition. Let $Z_0 = F_t$ denote the clean semantic target feature. In the forward process, Gaussian noise is progressively added to Z_0 over timesteps $t \in \{1, \dots, T\}$:

$$q(Z_t|Z_0) = \mathcal{N}(Z_t; \sqrt{\bar{\alpha}_t}Z_0, (1 - \bar{\alpha}_t)I), \quad (5)$$

where $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ and $\{\alpha_t\}$ follows a linear noise schedule. Equivalently,

$$Z_t = \sqrt{\bar{\alpha}_t}Z_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (6)$$

Table 1: Performance comparison of algorithms on VIREO Food-172, NUS-WIDE and MSRVT datasets. Metrics are Top-1/Top-5 Accuracy (Acc), Precision (Pre), and Recall (Rec). The best result of each metric is highlighted in bold.

Method	Model	VIREO Food-172		NUS-WIDE				MSRVT	
		Acc-1	Acc-5	Pre-1	Pre-5	Rec-1	Rec-5	Acc-1	Acc-5
Visual Modal Backbone	ResNet-18 (CVPR'16)	77.32	93.21	74.64	37.21	41.02	80.44	50.80	77.46
	WRN (BMVC'16)	82.29	95.52	78.73	39.36	44.01	85.33	49.57	79.93
	WiSeR (MMM'18)	82.76	96.51	78.90	39.45	44.14	85.52	50.80	81.60
	ViT-B/16 (ICLR'21)	85.37	97.29	80.46	40.57	45.50	87.96	53.25	81.85
	CMCRL (TIP'25)	86.80	97.16	82.62	41.30	46.95	89.45	54.36	82.71
Alignment Framework	CHAN (CVPR'23)	87.48	97.63	81.64	39.91	46.37	86.94	55.14	82.16
	HREM (CVPR'23)	88.12	96.98	81.31	39.36	46.14	85.68	54.76	83.66
	SWD (CVPR'19)	87.60	97.90	82.52	41.02	47.04	88.97	55.55	83.81
	SSAN (MM'20)	87.09	97.73	81.62	40.55	46.25	88.02	54.18	83.55
	CDD (CVPR'19)	86.01	96.97	81.60	40.52	46.25	87.96	54.68	81.84
	DAMM (CVPR'25)	88.19	97.27	82.36	40.97	46.73	89.11	56.18	83.31
	DiCA (AAAI'25)	87.78	97.96	83.17	41.35	47.36	89.61	55.15	84.58
	FMA (ICLR'26)	88.39	97.61	82.38	41.22	46.98	89.17	55.51	82.67
	RCDAlign	89.23	98.01	83.62	41.77	47.59	89.80	57.31	83.99

In the reverse process, the prediction network $\Phi_\theta(Z_t, t, F_v)$, equipped with RCAF and CGS, predicts the noise conditioned on the visual feature. The posterior distribution is written as:

$$q(Z_{t-1}|Z_t, Z_0) = \mathcal{N}(Z_{t-1}; \tilde{\mu}_t(Z_t, Z_0), \tilde{\beta}_t I). \quad (7)$$

Training minimizes the KL divergence between the true posterior q and the parameterized reverse distribution p_θ :

$$D_{\text{KL}}(q(Z_{t-1}|Z_t, Z_0) \| p_\theta(Z_{t-1}|Z_t, F_v)). \quad (8)$$

With reparameterization, this objective is simplified as the noise prediction loss:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t, Z_t, \epsilon} [\|\epsilon - \Phi_\theta(Z_t, t, F_v)\|_2^2]. \quad (9)$$

Thus, CFD performs progressive visual-to-semantic alignment instead of abrupt one-step feature matching.

4.2 Reliability-Consistency Adaptive Fusion

The noisy semantic state Z_t may contain uncertain or inconsistent textual information. Therefore, Reliability-Consistency Adaptive Fusion (RCAF) controls cross-modal interaction by jointly considering semantic reliability and visual-textual consistency. First, the uncertainty of Z_t is estimated by:

$$U_t = \sigma(M_{\text{unc}}(Z_t, t)), \quad U_t \in [0, 1]^D, \quad (10)$$

where M_{unc} is a two-layer MLP conditioned on the timestep embedding, and a larger U_t indicates lower reliability. Meanwhile, the semantic consistency between the visual feature and the noisy semantic state is computed as:

$$S_t = \cos(P_v F_v, P_t Z_t), \quad (11)$$

where P_v and P_t are learnable matrices projecting the two features into a shared interaction space. The reliability and consistency signals are then combined to generate the fusion gate:

$$G_t = \sigma(W_g[(1 - U_t); S_t \mathbf{1}]), \quad G_t \in [0, 1]^D, \quad (12)$$

where W_g is a learnable projection and $\mathbf{1}$ broadcasts the scalar consistency score to the feature dimension. In the cross-attention

operation, F_v is used as the Query, while Z_t serves as the Key and Value:

$$Q = F_v W_q, \quad K = Z_t W_k, \quad V = Z_t W_{val}. \quad (13)$$

The attention output is:

$$A_t = \text{Softmax}\left(\frac{QK^T}{\sqrt{D}}\right)V. \quad (14)$$

Finally, the gated fusion feature is obtained by:

$$Z_m = \text{LayerNorm}(F_v + G_t \odot A_t), \quad (15)$$

where $\odot$ denotes element-wise multiplication. In this way, RCAF enhances useful textual semantics while suppressing noisy or inconsistent guidance.

4.3 Confidence-Guided Selection

Although RCAF suppresses unreliable textual guidance, the fused feature Z_m may still be less discriminative than the original visual feature F_v . To address this issue, Confidence-Guided Selection (CGS) adaptively controls the contribution of Z_m according to feature divergence and prediction confidence. Given F_v and Z_m , we construct:

$$H_d = [F_v; Z_m; |F_v - Z_m|; F_v \odot Z_m]. \quad (16)$$

The divergence score is predicted as:

$$d = \sigma(\text{MLP}_d(H_d)), \quad d \in [0, 1]. \quad (17)$$

A smaller d indicates reliable fusion, while a larger d suggests stronger cross-modal conflict. The final representation is obtained by residual-style semantic correction:

$$Z_{\text{final}} = F_v + (1 - d) \cdot \gamma \cdot (Z_m - F_v), \quad (18)$$

where $\gamma = \sigma(\gamma_0)$ is a learnable scaling factor constrained to $[0, 1]$. Thus, the model adopts stronger fused semantics when d is small, and preserves the original visual feature when d is large. To supervise d without explicit conflict labels, we use prediction confidence. For single-label classification, the margin is defined as:

$$m_{\text{single}}(Z) = p_y(Z) - \max_{k \neq y} p_k(Z). \quad (19)$$

Table 2: Ablation study of RCDAlign.

Model	VIREO172		NUS-WIDE				MSRVTT	
	Acc-1	Acc-5	Pre-1	Pre-5	Rec-1	Rec-5	Acc-1	Acc-5
Baseline	85.37	97.29	80.46	40.57	45.50	87.96	53.25	81.85
+CFD	86.59	93.32	80.86	40.37	45.92	87.89	54.98	80.60
+CFD+RCAF	87.61	97.03	82.66	40.98	47.05	89.15	56.38	82.46
+CFD+RCAF+CGS	89.23	98.01	83.62	41.77	47.59	89.80	57.31	83.99

For multi-label classification, given the positive and negative label sets Y^+ and Y^- , the margin is:

$$m_{\text{multi}}(Z) = \frac{1}{|Y^+|} \sum_{y \in Y^+} p_y(Z) - \max_{k \in Y^-} p_k(Z). \quad (20)$$

We use $m(Z)$ to denote the corresponding margin under the current task. The pseudo-label and confidence weight are defined as:

$$e = \begin{cases} 1, & m(Z_m) > m(F_v), \\ 0, & \text{otherwise,} \end{cases} \quad (21)$$

$$w = |m(Z_m) - m(F_v)|. \quad (22)$$

The divergence feedback loss is:

$$\mathcal{L}_{dfl} = -\text{sg}(w) [\text{sg}(e) \log(1 - d + \epsilon) + (1 - \text{sg}(e)) \log(d + \epsilon)], \quad (23)$$

where ϵ ensures numerical stability and $\text{sg}(\cdot)$ denotes stop-gradient. During training, the margin is used only to supervise the divergence score. During inference, CGS directly predicts d and generates Z_{final} without ground-truth labels. The final representation is fed into the classifier, and the overall objective is:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_{diff} + \lambda_2 \mathcal{L}_{dfl}, \quad (24)$$

where $\mathcal{L}_{cls}$ is cross-entropy loss for the single-label datasets and binary cross-entropy for the multi-label dataset, and λ_1 and λ_2 balance the diffusion and selection objectives.

4.4 Inference Procedure

At test time the textual modality is unavailable, so the semantic state cannot be derived from F_t . Instead, RCDAlign starts from pure Gaussian noise $Z_T \sim \mathcal{N}(0, I)$ and performs conditional reverse denoising guided solely by the visual feature:

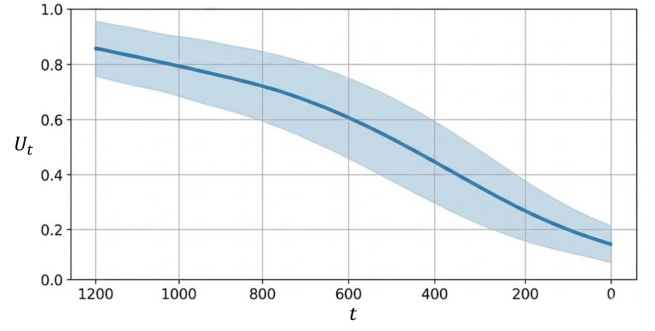
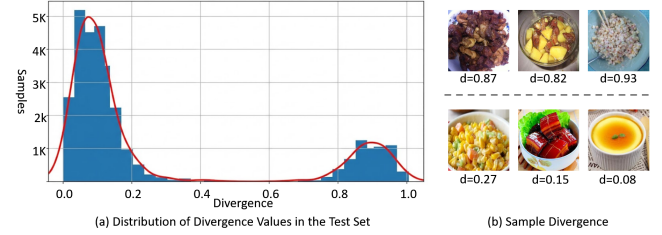
$$Z_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(Z_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \Phi_\theta(Z_t, t, F_v) \right) + \sigma_t \xi, \quad \xi \sim \mathcal{N}(0, I), \quad (25)$$

where Φ_θ internally applies RCAF at every step. After the denoising steps, the resulting semantic state yields Z_m , and CGS produces Z_{final} without access to any textual input or ground-truth label. Therefore textual information is used only to shape the target distribution during training and is never required at inference.

5 Experiment

5.1 Experiment Setting

5.1.1 Datasets. We evaluate RCDAlign on three public datasets: VIREO Food-172 [6] with 172 food categories, NUS-WIDE [12] with 81 multi-label semantic concepts, and MSRVTT [50] with 10,000 video clips and textual descriptions. These datasets cover diverse visual-textual scenarios for heterogeneous feature alignment.

**Figure 4: Change in uncertainty score during the denoising process.****Figure 5: Analysis of Confidence-Guided Selection.**

5.1.2 Evaluation Metrics. We use Top-1 and Top-5 accuracy for single-label classification, and Top-1 and Top-5 precision and recall for multi-label classification.

5.2 Performance Comparison

We compare RCDAlign with two groups of baselines. The first group includes visual backbone models, such as ResNet-18[19], WRN[53], WISer[34], ViT-B/16[15], and CMCRL[10]. The second group includes cross-modal alignment models, such as CHAN[35], HREM[17], SWD[27], SSAN[29], CDD[26], DAMM[56], DiCA[38], and FMA[24].

- As shown in Table 1, RCDAlign obtains the best results on seven of the eight reported metrics across the three datasets, demonstrating the effectiveness of progressive diffusion-based alignment. On MSRVTT Acc-5 it remains competitive with the strongest baseline.
- RCDAlign improves all Top-1 related metrics over the second-best method (VIREO172 Acc-1 +0.84, MSRVTT Acc-1 +1.13), indicating its ability to suppress noisy cross-modal interference and identify more discriminative categories.
- Compared with single-modal visual networks, RCDAlign consistently improves recognition performance, showing that textual privileged information can enhance visual representation learning when properly aligned.

5.3 Ablation Study

This section examines the performance of different modules based on ViT-B/16 in RCDAlign, with the results presented in Table 2.

- CFD improves all Top-1 metrics (VIREO172 +1.22, MSRVTT +1.73) but degrades Top-5 (VIREO172 Acc-5 97.29 $\rightarrow$ 93.32).

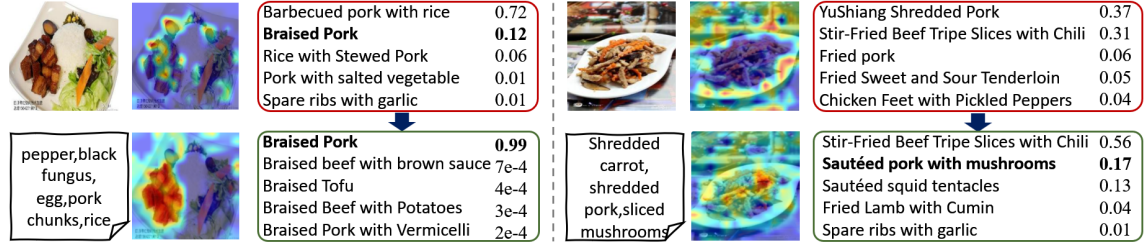


Figure 6: Comparison of Grad-CAM visualizations.

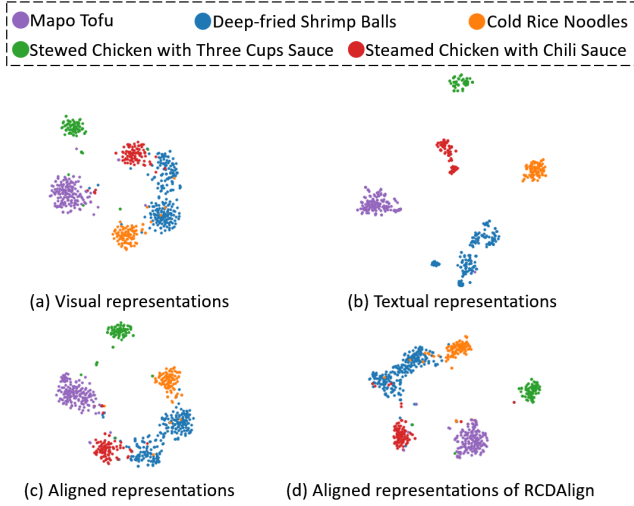


Figure 7: T-SNE visualizations of different feature alignment methods.

We attribute this to the mode-seeking nature of unconstrained denoising: uniform textual guidance over-commits features to a single semantic mode, sharpening the top prediction while distorting the calibration of remaining candidates. This motivates an explicit reliability mechanism.

- RCAF recovers and surpasses the lost Top-5 (93.32 $\rightarrow$ 97.03) and further improves Top-1 (NUS-WIDE Rec-1 +1.13), confirming that uncertainty- and consistency-driven gating suppresses unreliable guidance rather than merely rescaling it.
- The full model performs best overall. As CGS operates on RCAF’s output Z_m , the two modules are sequentially dependent rather than interchangeable.

5.4 In-depth Analysis

5.4.1 Analysis of Reliability-Consistency Adaptive Fusion. To further analyze RCAF, we report the average uncertainty score U_t during reverse denoising in Fig. 4. At early stages, strong noise leads to higher uncertainty, so RCAF suppresses unreliable textual guidance. As denoising proceeds, semantic information becomes clearer and U_t gradually decreases, allowing stronger textual information to be fused. This trend shows that RCAF adaptively controls fusion strength according to feature reliability and improves alignment robustness.

5.4.2 Analysis of Confidence-Guided Selection. To evaluate CGS, we record the divergence score d on the test set. As shown in Fig. 5(a), d presents a bimodal distribution, where low-divergence samples indicate reliable fusion and high-divergence samples suggest semantic conflicts. Fig. 5(b) further shows that high-divergence samples often correspond to blur, poor illumination, or complex ingredients. These results indicate that CGS can detect unreliable fused features and reduce negative transfer.

5.5 Case Study

5.5.1 Grad-CAM Analysis. We also use Grad-CAM to analyze the attention regions of the baseline and RCDAlign on challenging food images, as shown in Fig. 6. For images with complex ingredients, the baseline tends to focus on background regions or non-discriminative side dishes, leading to incorrect predictions. In contrast, RCDAlign can better locate the key food regions and suppress background interference. This indicates that the proposed method helps the model extract more discriminative visual evidence under complex visual conditions.

5.5.2 T-SNE Visualization of Different Alignment Methods. We visualize feature distributions on VIREO Food-172 using T-SNE, selecting five categories with poor baseline performance. As shown in Fig. 7, traditional alignment methods only slightly improve category separation, while RCDAlign produces more compact intra-class clusters and clearer inter-class boundaries, demonstrating its ability to enhance feature discriminability.

6 Conclusion

In this paper, we propose **RCDAlign**, a Reliability-Consistency Guided Diffusion Alignment framework that reformulates text-guided visual representation learning as a conditional denoising process. By integrating Conditional Feature Diffusion, Reliability-Consistency Adaptive Fusion, and Confidence-Guided Selection, RCDAlign progressively aligns visual features with textual semantics while suppressing noisy guidance and reducing negative transfer. Experiments on three public datasets demonstrate that RCDAlign improves visual representation discriminability and achieves competitive recognition performance. A main limitation is the additional inference cost introduced by iterative denoising; reducing the number of sampling steps without degrading alignment quality remains an open problem. In future work, we plan to extend RCDAlign to broader privileged modalities such as audio and structured knowledge.

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